

Linear Regression Questions



Goal: Cover math, intuition, implementation, debugging, and real-world thinking.

Fundamental Questions (Core Understanding)

1. What assumptions does Linear Regression make about the data?
2. Explain the difference between correlation and regression.
3. What is the cost function used in Linear Regression? Why?
4. What is the difference between OLS and Gradient Descent solution?
5. What is multicollinearity and why is it dangerous?
6. What is the bias-variance tradeoff in context of Linear Regression?
7. How does Ridge regression modify the cost function?
8. How does Lasso perform feature selection?
9. When would ElasticNet be preferred over Ridge and Lasso?
10. Why should we standardize features before applying regularization?

Breadth First Questions (Concept Coverage Across Scenarios)

1. How does Linear Regression behave when outliers exist?
2. How do you evaluate a regression model? Compare MAE, MSE, RMSE, R^2 .
3. What happens if features are highly correlated?
4. What if number of features > number of samples?
5. How does scaling affect Ridge vs Lasso?
6. What happens when $\alpha \rightarrow 0$ in Ridge?
7. What happens when $\alpha \rightarrow \infty$ in Lasso?
8. Can Linear Regression be used for classification? Why or why not?
9. How would you handle categorical variables in regression?
10. How do you detect overfitting in regression?

Depth First Questions (Deep Technical Understanding)

1. Derive the normal equation for Linear Regression.
2. Prove why OLS estimator is unbiased.
3. Why does L1 regularization produce sparse weights?
4. Explain geometrically how Ridge regression shrinks coefficients.
5. Show mathematically how multicollinearity increases variance.
6. Why does MSE amplify large errors more than MAE?
7. Compare coordinate descent vs gradient descent for Lasso.
8. Explain convexity of regression cost function.
9. When does closed-form solution fail numerically?
10. How does condition number relate to stability?

Case Study Questions (Real-World Thinking)

1. You're predicting house prices and R^2 is 0.92 on train but 0.61 on test. What do you do?
2. A dataset has 500 features and 200 samples. Which regression variant do you choose and why?
3. Business wants interpretability more than accuracy. Which model and why?
4. Your residuals show a pattern. What does that indicate?
5. Outliers heavily influence your coefficients. What practical steps would you take?
6. Client wants feature importance ranking. How would you provide it?
7. Data contains strong multicollinearity. Explain your solution.
8. Model predictions are biased low for high values. Why might that happen?
9. You deploy model and data distribution shifts. What now?
10. You are building salary prediction. Ethical risks?



Logistic Regression Questions



Goal: Cover math, intuition, implementation, debugging, and real-world thinking.

Fundamental Questions (Core Understanding)

1. Why can't we use Linear Regression for binary classification?
2. What is the sigmoid function and why is it used?
3. What is the cost function of Logistic Regression?
4. Why don't we use MSE for Logistic Regression?
5. What does the output of Logistic Regression represent?
6. What is the decision boundary?
7. What assumptions does Logistic Regression make?
8. What is odds and log-odds?
9. How does regularization work in Logistic Regression?
10. What is the difference between L1 and L2 in classification context?

Breadth First Questions (Concept Coverage Across Scenarios)

1. How do you evaluate Logistic Regression? Compare accuracy, precision, recall, F1, ROC-AUC.
2. What happens if classes are imbalanced?
3. How do you handle multicollinearity in Logistic Regression?
4. How do you interpret coefficients?
5. What happens if features are not scaled?
6. What is the effect of increasing regularization strength?
7. Can Logistic Regression handle multi-class classification?
8. What is the difference between one-vs-rest and softmax?
9. How do you choose threshold other than 0.5?
10. When would Logistic Regression outperform complex models?

Depth First Questions (Deep Technical Understanding)

1. Derive the log-likelihood function for Logistic Regression.
2. Why is the cost function convex?
3. Why does Logistic Regression not have a closed-form solution?
4. Show how gradient descent updates weights.
5. Why does L1 regularization create sparse solutions?
6. Explain the connection between Logistic Regression and maximum likelihood estimation.
7. Prove why sigmoid output is bounded between 0 and 1.
8. How does decision boundary relate to linear equation?
9. What happens when data is perfectly separable?
10. Explain Hessian matrix role in Newton's method for Logistic Regression?

Case Study Questions (Real-World Thinking)

1. You are predicting loan default. False negatives are very costly. What metric do you optimize?
2. Model accuracy is 95% but recall for positive class is 40%. Explain.
3. Dataset has 2% fraud cases. How do you train?
4. Coefficients are unstable across training runs. Why?
5. Business wants probability, not just classification. How do you present it?
6. ROC-AUC is good but business says predictions are bad. Why?
7. Model performs well in training but degrades after deployment. What happened?
8. Feature coefficients are very large. What does that indicate?
9. You need explainability for legal reasons. Why is Logistic Regression useful?
10. You have 1M rows and 20 features. Would you use Logistic Regression or a deep model?





Decision Tree Questions

Goal: Cover math, intuition, implementation, debugging, and real-world thinking.

Fundamental Questions (Core Understanding)

1. What is a Decision Tree and how does it work?
2. What is entropy?
3. What is Gini impurity?
4. Difference between entropy and Gini?
5. What is information gain?
6. How does a Decision Tree choose the best split?
7. What causes overfitting in Decision Trees?
8. What is tree depth?
9. Difference between classification tree and regression tree?
10. Why don't Decision Trees require feature scaling?

Breadth First Questions (Concept Coverage Across Scenarios)

1. How do you prevent overfitting in a Decision Tree?
2. What is pruning? Difference between pre-pruning and post-pruning?
3. What happens if the dataset has noise?
4. How do Decision Trees handle categorical features?
5. How do they handle missing values?
6. Why are Decision Trees unstable?
7. What happens when features are highly correlated?
8. How do you evaluate a Decision Tree model?
9. When would you prefer Decision Tree over Logistic Regression?
10. What are limitations of Decision Trees?

Depth First Questions (Deep Technical Understanding)

1. Derive entropy formula.
2. Why is entropy maximum at $p = 0.5$?
3. Show how Gini impurity behaves mathematically.
4. Prove why Information Gain is always non-negative.
5. Explain computational complexity of building a Decision Tree.
6. Why are Decision Trees considered greedy algorithms?
7. What is bias-variance behavior of a deep tree?
8. How does split threshold selection work for continuous variables?
9. Why do small changes in data drastically change the tree?
10. How does CART algorithm differ from ID3?

Case Study Questions (Real-World Thinking)

1. Your Decision Tree has 100% training accuracy but 65% test accuracy. What happened?
2. Business wants interpretable rules. Why is Decision Tree good?
3. Tree depth is 20 and performance unstable. What would you tune?
4. You have 50 features but only 200 samples. What risks exist?
5. Dataset has severe class imbalance. What do you adjust?
6. Model is too sensitive to small data changes. What solution?
7. You are predicting credit approval. What ethical risks exist?
8. How would you explain feature importance from tree?
9. When would you switch from Decision Tree to Random Forest?
10. In production, prediction time must be very fast. Is Decision Tree suitable?



Random Forest Questions



Goal: Cover math, intuition, implementation, debugging, and real-world thinking.

Fundamental Questions (Core Understanding)

1. What is a Random Forest?
2. What problem of Decision Trees does Random Forest solve?
3. What is bagging?
4. What is bootstrap sampling?
5. Why does Random Forest select random subsets of features?
6. How does averaging reduce variance?
7. Difference between Random Forest classifier and regressor?
8. What is out-of-bag (OOB) error?
9. Why is Random Forest less prone to overfitting than a single tree?
10. Why does Random Forest not require feature scaling?

Breadth First Questions (Concept Coverage Across Scenarios)

1. How do you tune a Random Forest model?
2. What happens if `n_estimators` is very small?
3. What happens if `max_depth` is very large?
4. How does Random Forest handle missing values?
5. How does it deal with imbalanced data?
6. How do you evaluate feature importance?
7. When would Random Forest outperform Logistic Regression?
8. When would it fail compared to boosting models?
9. How does Random Forest behave with noisy data?
10. Is Random Forest suitable for high-dimensional sparse data?

Depth First Questions (Deep Technical Understanding)

1. Explain mathematically how bagging reduces variance.
2. Why does random feature selection decorrelate trees?
3. What is the bias-variance tradeoff in Random Forest?
4. Derive expected prediction of a Random Forest regressor.
5. Why is Random Forest still a greedy algorithm at tree level?
6. Explain time complexity of training Random Forest.
7. How does OOB estimation work internally?
8. Why does increasing trees not increase overfitting much?
9. Explain difference between bagging and boosting mathematically.
10. How does Random Forest approximate a non-linear function?

Case Study Questions (Real-World Thinking)

1. Your Random Forest has 99% training accuracy and 96% test accuracy. Is that good or suspicious?
2. Feature importance shows one variable dominating. What could be wrong?
3. Model is too slow in production. What parameters would you change?
4. Dataset has 1 million rows and 500 features. Is Random Forest appropriate?
5. You need model explainability for regulators. Is Random Forest ideal?
6. Business wants probability calibration. What will you check?
7. Model performance drops after deployment. Possible reasons?
8. You have extremely imbalanced fraud data. What would you modify?
9. Training time is too long. Practical optimization strategies?
10. When would you choose Gradient Boosting over Random Forest?



Gradient Boosting Questions



Goal: Cover math, intuition, implementation, debugging, and real-world thinking.

Fundamental Questions (Core Understanding)

1. What is Gradient Boosting?
2. How is boosting different from bagging?
3. What problem does boosting solve that Random Forest does not?
4. What is a weak learner in Gradient Boosting?
5. What does each new tree learn?
6. What is the role of learning rate?
7. What is the loss function in regression boosting?
8. How does boosting work for classification?
9. Why is Gradient Boosting sequential?
10. Why is it more prone to overfitting than Random Forest?

Breadth First Questions (Concept Coverage Across Scenarios)

1. How do you tune `learning_rate` and `n_estimators` together?
2. What happens if learning rate is very high?
3. What happens if learning rate is very low?
4. What is subsampling in boosting?
5. How does Gradient Boosting handle outliers?
6. How does it perform with noisy data?
7. What is the impact of tree depth?
8. When would you prefer Gradient Boosting over Random Forest?
9. Why is boosting slower than bagging?
10. What is early stopping and why is it important?

Depth First Questions (Deep Technical Understanding)

1. Explain how boosting is gradient descent in function space.
2. Derive update step for Gradient Boosting regression.
3. Why do we fit residuals in each step?
4. How is classification boosting linked to logistic loss?
5. Explain shrinkage mathematically.
6. Why does boosting reduce bias?
7. Show bias-variance behavior of boosting.
8. What happens when trees are too deep?
9. Compare AdaBoost vs Gradient Boosting mathematically.
10. Why does boosting minimize differentiable loss functions?

Case Study Questions (Real-World Thinking)

1. Model performs extremely well on training but poorly on test. What adjustments?
2. Dataset has 5% fraud cases. How do you tune for this?
3. Training takes too long. What strategies reduce time?
4. Feature importance is unstable across runs. Why?
5. You must deploy on low-latency system. Is GBDT ideal?
6. Model performance improves as trees increase, then suddenly degrades. Why?
7. You need calibrated probabilities. What do you check?
8. Business wants explanation of prediction. How do you provide it?
9. Data distribution shifts after deployment. What risks exist?
10. When would you choose XGBoost/LightGBM instead of vanilla Gradient Boosting?



K-Means Clustering Questions



Goal: Cover math, intuition, implementation, debugging, and real-world thinking.

Fundamental Questions (Core Understanding)

1. What is K-Means clustering?
2. What is the objective function of K-Means?
3. What does "minimizing within-cluster variance" mean?
4. What are centroids?
5. Why is Euclidean distance commonly used?
6. How does the algorithm initialize clusters?
7. What is convergence in K-Means?
8. Why must we choose K beforehand?
9. Does K-Means guarantee global optimum?
10. Why is feature scaling important in K-Means?

Breadth First Questions (Concept Coverage Across Scenarios)

1. How do you choose the optimal value of K?
2. What is the Elbow Method?
3. What is Silhouette Score?
4. How does K-Means behave with outliers?
5. Can K-Means handle categorical data?
6. What happens if clusters are not spherical?
7. What happens if clusters have different densities?
8. How does initialization affect results?
9. What is K-Means++?
10. When would you not use K-Means?

Depth First Questions (Deep Technical Understanding)

1. Write the mathematical objective function of K-Means.
2. Why does K-Means converge in finite steps?
3. Is K-Means convex optimization? Why or why not?
4. Explain time complexity of K-Means.
5. Why is it sensitive to initialization?
6. Prove that each iteration reduces the objective function.
7. Why does K-Means prefer spherical clusters?
8. How does high dimensionality affect distance computation?
9. Explain relationship between K-Means and EM algorithm.
10. How does Mini-Batch K-Means work?

Case Study Questions (Real-World Thinking)

1. You cluster customers but business says clusters are meaningless. What went wrong?
2. Dataset has extreme outliers. What would you do?
3. You scaled features incorrectly. What happens?
4. Clusters change every time you run the model. Why?
5. Data has 100 features. What risks exist?
6. You don't know how many clusters exist. What approach?
7. Clusters overlap heavily. Is K-Means appropriate?
8. You must cluster millions of records. What optimization?
9. How would you validate clusters without labels?
10. When would you use DBSCAN instead of K-Means?



PCA Questions



Goal: Cover math, intuition, implementation, debugging, and real-world thinking.

Fundamental Questions (Core Understanding)

1. What is PCA?
2. What problem does PCA solve?
3. What is variance in the context of PCA?
4. What is a principal component?
5. Why are principal components orthogonal?
6. What does eigenvalue represent in PCA?
7. What does eigenvector represent?
8. Why do we standardize features before PCA?
9. Is PCA supervised or unsupervised?
10. Does PCA reduce noise? Why or why not?

Breadth First Questions (Concept Coverage Across Scenarios)

1. How do you decide number of principal components?
2. What is explained variance ratio?
3. What happens if features are not scaled?
4. How does PCA behave with correlated features?
5. Can PCA handle categorical data?
6. What happens in high-dimensional datasets?
7. When would PCA harm model performance?
8. How is PCA different from feature selection?
9. Can PCA improve model training speed?
10. When would you not use PCA?

Depth First Questions (Deep Technical Understanding)

1. Derive PCA using covariance matrix.
2. Why do we compute eigen decomposition of covariance matrix?
3. Show the mathematical objective PCA optimizes.
4. Why does first principal component maximize variance?
5. Prove orthogonality of principal components.
6. Explain relationship between PCA and SVD.
7. What is reconstruction error?
8. Why is PCA a linear transformation?
9. Is PCA a convex optimization problem?
10. How does dimensionality reduction affect bias-variance tradeoff?

Case Study Questions (Real-World Thinking)

1. You reduced 100 features to 5 PCs and model accuracy dropped. Why?
2. Business wants interpretability. Is PCA good?
3. Dataset has 95% variance explained by first 2 components. What does that mean?
4. You apply PCA before Logistic Regression. Why might it help?
5. You apply PCA before Random Forest. Is it necessary?
6. Data is sparse and high-dimensional. Should you use PCA?
7. PCA components are hard to explain. How do you present results?
8. You suspect multicollinearity. Can PCA help?
9. You perform PCA on training + test together. What's wrong?
10. When would you use Kernel PCA instead?



PCA Questions

